AI-Powered Data Analytics Internship

Week 3 Submission:

Trained Machine Learning Model, Predicted Probabilities and Visual Insights

Date Submitted:

3rd August 2026

Submitted by:

Team 02

Team Member Name	Email Address
Inshaal Tariq	inshaaltariq774@gmail.com
Noor ul Ain Zahid	noorulainzahid289@gmail.com
Suchana Panja	suchanapanja7@gmail.com
Adeniji Shukrat Opeyemi	adenijishukratopey@gmail.com
Simphiwe Mdakane	simphiwemdakane9@gmail.com
Gideon Asante Yeboah	gasante099@gmail.com
Shivanshi Agrawal	emailatshivanshi@gmail.com
Oladeji Faith	oladejifaith552@gmail.com
Tayyaba Munir	bsitf23m08@bgnu.edu.pk
Soumya Ranjan Jena	soumyajena2005@gmail.com
Chabi Raleting	chabi21@gmail.com

Executive Summary

This report documents the complete Week 3 workflow of the AI-Powered Data Analytics Internship project: the development, evaluation, and comparison of a machine learning model built to predict whether a learner's application to a given opportunity will result in active participation. The work builds directly on the cleaned dataset and engineered features produced in Week 2, and extends that foundation into a fully trained, validated, and interpretable predictive system.

Using a dataset of 8,558 learner-opportunity records, the team constructed a thirteen-feature, leakage-free predictor set, trained a Random Forest classifier, and benchmarked it against three alternative algorithms: Gradient Boosting, a Decision Tree, and Logistic Regression. The final Random Forest model achieved a test accuracy of 85.69%, a precision of 90.48%, a recall of 78.16%, an F1-score of 83.87%, and a ROC-AUC of 0.9225, with a train-test accuracy gap of only 1.43 percentage points, indicating low overfitting risk and strong generalisation to unseen data.

Beyond raw performance, the model was used to generate calibrated participation probabilities for every opportunity on the platform, which were then ranked, validated against historical outcomes, and visualised through a series of charts and heatmaps. The single most important finding to emerge from this analysis is what the team terms the category-volume paradox: opportunities in the Internship category attract by far the largest number of applications, yet consistently show the lowest predicted participation probabilities (14.5%-31.5%), while lower-volume categories such as Events, Competitions, and Courses show probabilities as high as 92%-97.5%. This single insight, together with the supporting feature importance and comparison analysis presented in this report, gives programme managers a concrete, evidence-based basis for prioritising opportunities and allocating engagement resources ahead of Week 4.

Contents

1. Introduction and Objectives

The overarching goal of this project is to help programme managers understand, quantify, and act on the factors that drive active learner participation. Week 1 established a cleaned dataset from the raw RIT/Riipen learner signup and application records. Week 2 built on that foundation with exploratory cohort analysis, derived feature engineering, and early forecasting signals. Week 3, documented in this report, converts those signals into an operational predictive system.

The specific objectives for this phase of the project were to:

- Re-verify the Week 2 preprocessed dataset and confirm it is suitable for supervised machine learning.
- Engineer a final, leakage-free feature set and select an appropriate target variable for predicting active participation.
- Train a primary classification model, document its architecture and hyperparameters, and rigorously evaluate its performance.
- Compare the primary model against at least two alternative algorithms to justify the final model selection with evidence rather than assumption.
- Generate calibrated participation probabilities for every opportunity and validate those probabilities against historical outcomes.
- Produce a complete set of visualisations, feature importance rankings, heatmaps, and a confusion matrix to communicate model behaviour to non-technical stakeholders.
- Translate the technical results into concrete, actionable recommendations for programme managers heading into Week 4.

Each of the following sections documents one stage of this pipeline in the order it was actually executed, so that a reader unfamiliar with the underlying notebooks can follow the complete reasoning from raw data to final recommendation.

2. Model Training and Performance

2.1 Dataset Readiness

The modelling dataset originates from the Week 2 preprocessed export of the RIT learner signup and application platform. It contains 8,558 individual learner application records across 34 raw columns, spanning demographic fields (gender, country, institution, intended major), opportunity metadata (category, start and end dates), timing fields (signup date, apply date, entry creation date), and outcome fields (status code, status description, and two derived participation flags).

The target variable for this project is is_active_participation, a binary indicator derived in Week 2 from the raw status_description field. A value of 1 indicates the learner reached an active participation outcome (for example, Team Allocated, Started, or Rewards Award); a value of 0 indicates the application did not progress to active participation (for example, Rejected, Withdrawn, or Dropped Out). The class balance is close to even, with 4,486 negative cases (52.42%) and 4,072 positive cases (47.58%), which means the model does not require aggressive resampling to avoid a majority-class bias, although class_weight="balanced" was still applied during training as an additional safeguard.

Table 1: Dataset overview.

Property	Value
Total records	8,558
Raw columns	34
Target variable	is_active_participation
Positive class (Active = 1)	4,072 records (47.58%)
Negative class (Not Active = 0)	4,486 records (52.42%)
Train / Test split	6,846 / 1,712 records (80% / 20%, stratified)

Every column was audited for missing values before feature engineering began. Several columns showed high missingness, especially engagement_duration_days (5,085 missing, 59.4%) and opportunity_start_date (4,637 missing, 54.2%). These fields are only populated after a learner begins engaging with an opportunity, indicating a structural data characteristic rather than a quality issue, which later influenced feature selection.

- opportunity_start_date: 4,637 missing (54.2%)
- opportunity_end_date: 1,262 missing (14.7%)
- engagement_duration_days: 5,085 missing (59.4%)
- days_to_apply: 534 missing (6.2%)
- learner_signup_datetime / apply_date / signup_year / signup_month / signup_quarter: 295-307 missing each (approximately 3.4%)

Two imputation strategies were applied based on data type. Numerical features (such as age and days_to_apply) had missing values replaced with the column median, as it is more robust to outliers and skewed behavioural data. Categorical features (such as gender and apply_speed_category) were filled with the label "Unknown" instead of the mode, allowing the model to learn missingness patterns rather than merging missing values with the majority category.

Because Random Forest, Gradient Boosting, and Decision Tree models split on individual feature thresholds rather than assuming linear relationships, Label Encoding was used to convert each categorical column into an integer representation, which a tree can recover any arbitrary grouping from through repeated splits. For the Logistic Regression baseline, the same encoded features were passed through StandardScaler because it is sensitive to feature scale. The country field was handled separately by keeping the ten most frequent countries individually and grouping the remaining values into an "Other" category, creating a stable eleven-level feature.

An initial candidate set of 15 features was assembled from the Week 2 engineered columns. Before training, each feature was checked for data leakage. Two features, **engagement_duration_days** and **duration_category**, were found to leak information because they are only known after learner participation occurs, so both were removed. **total_category_opportunities** was also examined due to its strong correlation with the target but was retained, as it is determined before the outcome and represents a valid structural characteristic rather than leaked information.

The final, leakage-free feature set used for all modelling in this report consists of thirteen variables, summarised in Table 2 below.

Table 2: Final feature set used for model training.

#	Feature	Type	Description
1	gender	Categorical	Learner-reported gender
2	country_grouped	Categorical	Top 10 countries by frequency, remainder grouped as "Other"
3	opportunity_category	Categorical	Internship, Course, Event, Competition, or Engagement
4	apply_weekday	Categorical	Day of the week the application was submitted
5	apply_speed_category	Categorical	Bucketed speed of application after signup
6	apply_month	Numerical	Calendar month of application
7	apply_quarter	Numerical	Academic/calendar quarter of application
8	is_weekend	Binary	Whether the application was submitted on a weekend
9	age	Numerical	Learner's age at time of application
10	days_to_apply	Numerical	Days elapsed between platform signup and application
11	participation_frequency	Numerical	Learner's historical participation count
12	is_repeat_participant	Binary	Whether the learner has participated before
13	total_category_opportunities	Numerical	Fixed count of opportunities available in that category

Figure 1 shows the resulting target class distribution used for training and evaluation.

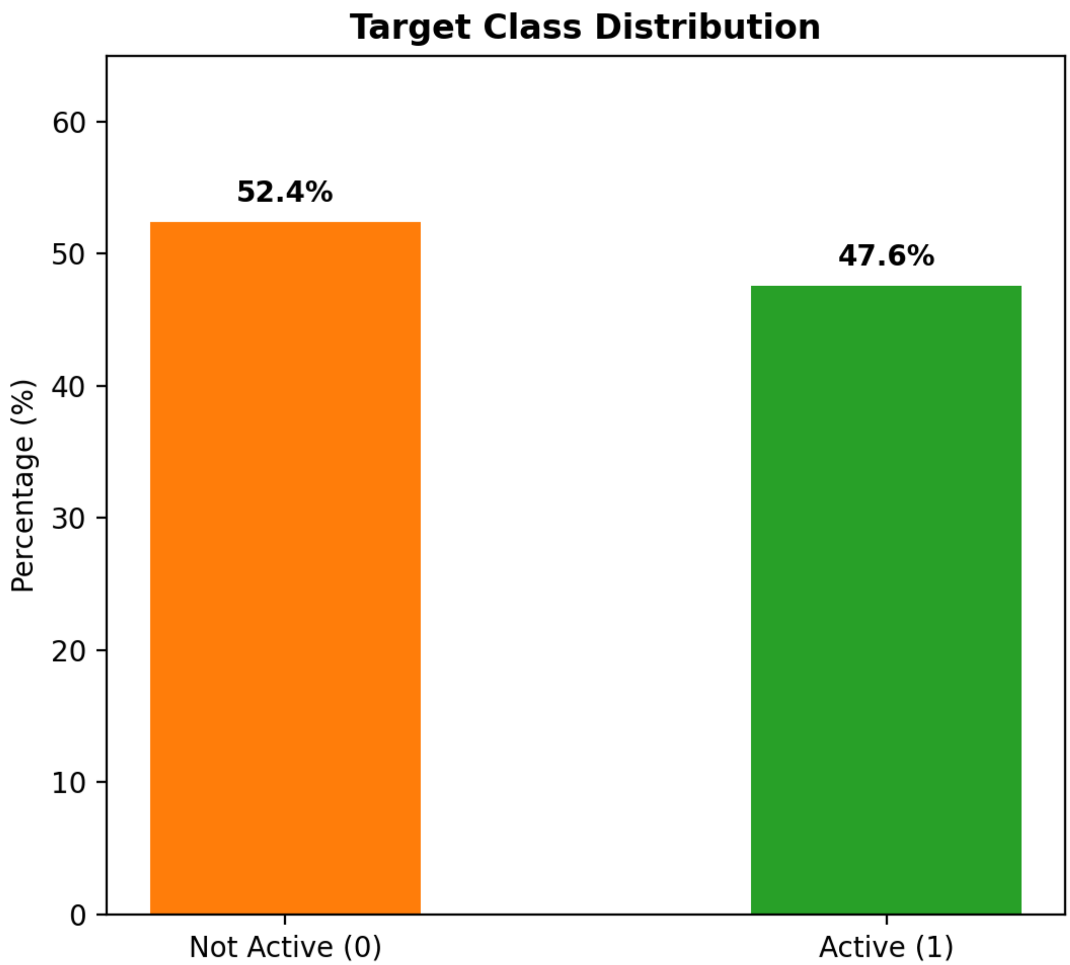

Figure 1: Target class distribution: Active versus Not Active learners in the dataset.

At 52.42% Not Active versus 47.58% Active, the dataset is close enough to balanced that the model does not require aggressive resampling techniques, though class_weight="balanced" was retained as an additional safeguard during training.

2.2 Model Building

The dataset was split into training and testing partitions using an 80/20 ratio (6,846 training records and 1,712 test records), with the split stratified on the target variable to preserve class proportions. A fixed random_state of 42 was used throughout the project to ensure that all models, charts, and metrics are reproducible. Instead of selecting one algorithm initially, the team compared multiple models using the same feature set, train/test split, and evaluation metrics, selecting the best model based on quantitative results detailed in Section 2.4.

The primary model selected for this project is a Random Forest Classifier, an ensemble method that combines multiple decision trees trained on bootstrapped subsets of data. It was chosen because it handles mixed feature types without scaling, reduces overfitting through random sampling of rows and features, and provides feature importance scores needed for model interpretation in Sections 4 and 5.

Table 3: Random Forest hyperparameter configuration.

Parameter	Value	Purpose
n_estimators	300	Number of trees in the ensemble; large enough for stable, low-variance predictions
max_depth	10	Caps tree complexity to reduce overfitting risk
min_samples_leaf	5	Prevents very small, noise-driven leaf splits
class_weight	balanced	Adds robustness against the slight class imbalance (52.4% / 47.6%)
random_state	42	Fixes randomness for full reproducibility
n_jobs	-1	Uses all available CPU cores to parallelise training

The model was fit on the 6,846-record training partition using the thirteen label-encoded features described in Section 2.1. Training completed in approximately 1.2-1.4 seconds on the available hardware, which is a practical consideration for a project where the model may need to be periodically retrained as new application data arrives.

2.3 Model Evaluation

The trained model was evaluated on the 1,712-record held-out test set, which the model had not seen during training. The five standard classification metrics below were computed to give a complete picture of model quality from multiple angles: overall correctness (accuracy), the cost of false alarms (precision), the cost of missed positives (recall), a balance of the two (F1-score), and the model's ability to rank positive cases above negative cases across all possible thresholds (ROC-AUC).

Table 4: Random Forest performance on the held-out test set.

Metric	Score	Interpretation
Train Accuracy	87.12%	Accuracy on data the model was trained on
Test Accuracy	85.69%	Accuracy on unseen data
Precision	90.48%	Of learners predicted "Active", 90.48% actually were
Recall	78.16%	Of learners who were actually active, 78.16% were correctly identified

F1-Score	83.87%	Harmonic mean of precision and recall
ROC-AUC	0.9225	Strong ability to rank active learners above inactive learners
Train-Test Gap	1.43 pts	Low overfitting risk (well under the 7-point caution threshold)

The precision-recall balance shows that the model is intentionally conservative. It rarely classifies inactive learners as active (high precision), but it also misses around one in five actual participants (78.16% recall). For programme prioritisation, this trade-off is acceptable because avoiding false predictions is often more valuable than missing potential participants identified through other signals.

A model that performs better on training data than unseen test data is said to be overfitting: it has memorised parts of the training set instead of learning generalisable patterns. Figure 2 compares both accuracy scores directly.

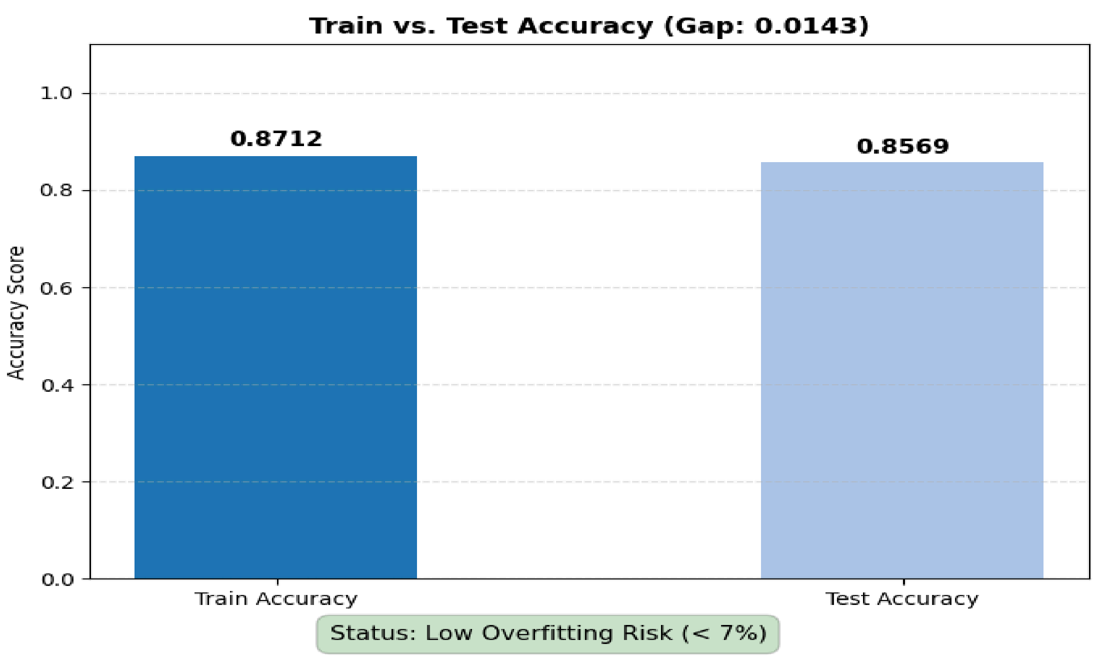

Figure 2: Training accuracy versus testing accuracy for the Random Forest model.

The gap between the two bars is only 1.43 percentage points (87.12% train versus 85.69% test), which falls well inside the conventional 7-point threshold used to flag possible overfitting. This confirms the max_depth=10 and min_samples_leaf=5 constraints applied in Section 2.2 were effective at controlling model complexity.

2.4 Model Comparison

To ensure the Random Forest model was not selected by default, it was benchmarked against three alternative algorithms, each trained and evaluated on the identical leakage-free feature set and train/test split described in Section 2.1 and Section 2.2. Four algorithms were evaluated in total:

- Random Forest Classifier - an ensemble of decision trees trained on bootstrapped samples, chosen as the primary candidate for the reasons discussed above.
- Gradient Boosting Classifier - an ensemble method that builds trees sequentially, with each new tree correcting the errors of the previous ones. It typically achieves strong accuracy but trains more slowly than Random Forest because trees cannot be built in parallel.
- Decision Tree Classifier - a single tree used as a simplicity baseline, to quantify how much benefit the ensemble methods provide over a single, easily-interpretable model.
- Logistic Regression - a linear baseline that models the log-odds of active participation as a weighted sum of the input features. Included as a baseline because it is fast, well-understood, and provides a useful lower bound for comparison against the more flexible tree-based methods.

All four models were trained with a fixed random_state=42 on the same 6,846-record training partition and evaluated on the same 1,712-record test partition. Table 5 and Figure 3 summarise the results, ranked by ROC-AUC.

Table 5: Comparative performance of all four evaluated models.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Train Time (s)
Random Forest	85.69%	90.48%	78.16%	83.87%	0.9225	1.36
Gradient Boosting	85.86%	90.64%	78.40%	84.08%	0.9221	1.52
Decision Tree	84.87%	87.37%	79.75%	83.39%	0.9138	0.02
Logistic Regression	82.71%	87.88%	73.87%	80.27%	0.8923	0.02

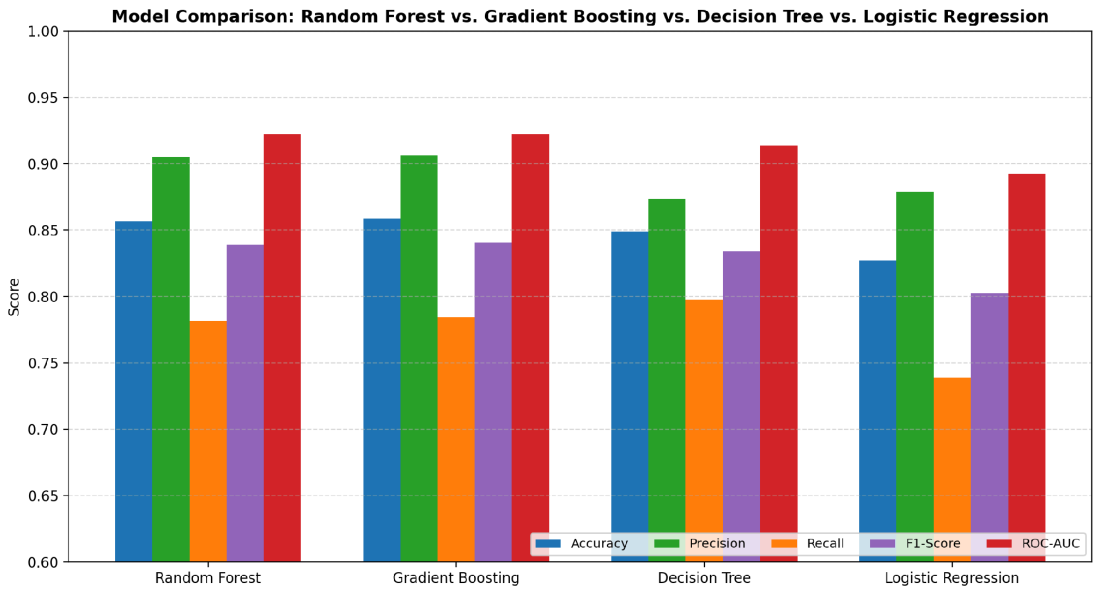

Figure 3: Comparison of all four models across accuracy, precision, recall, F1-score, and ROC-AUC.

The results show a clear two-tier structure. Random Forest and Gradient Boosting form the top tier, outperforming Decision Tree and Logistic Regression across all metrics. However, the two top models are nearly identical, with ROC-AUC scores differing by only 0.0004 (0.9225 vs. 0.9221), and Gradient Boosting's slightly higher accuracy (85.86% vs. 85.69%) is likely due to normal sampling variation.

Since both models performed similarly, the final selection was based on practical factors. Random Forest trained faster (1.36 s vs. 1.52 s), requires fewer hyperparameters, and has lower overfitting risk. Therefore, it was selected as the final model, while Gradient Boosting remains a strong alternative for future larger datasets.

3. Predicted Probabilities

3.1 Opportunity Probability vs. Application Volume

Beyond the binary Active / Not Active classification evaluated in Section 2.3, the trained Random Forest model was used in its probabilistic mode (predict_proba) to generate a continuous participation probability, between 0 and 1, for every opportunity on the platform. Because a single opportunity typically has multiple learner-level records, probabilities were first predicted at the individual learner-application level and then averaged up to the opportunity level, producing one representative predicted probability per opportunity. Figure **4** plots the average predicted probability against the total number of applications, with opportunities coloured according to their assigned priority level.

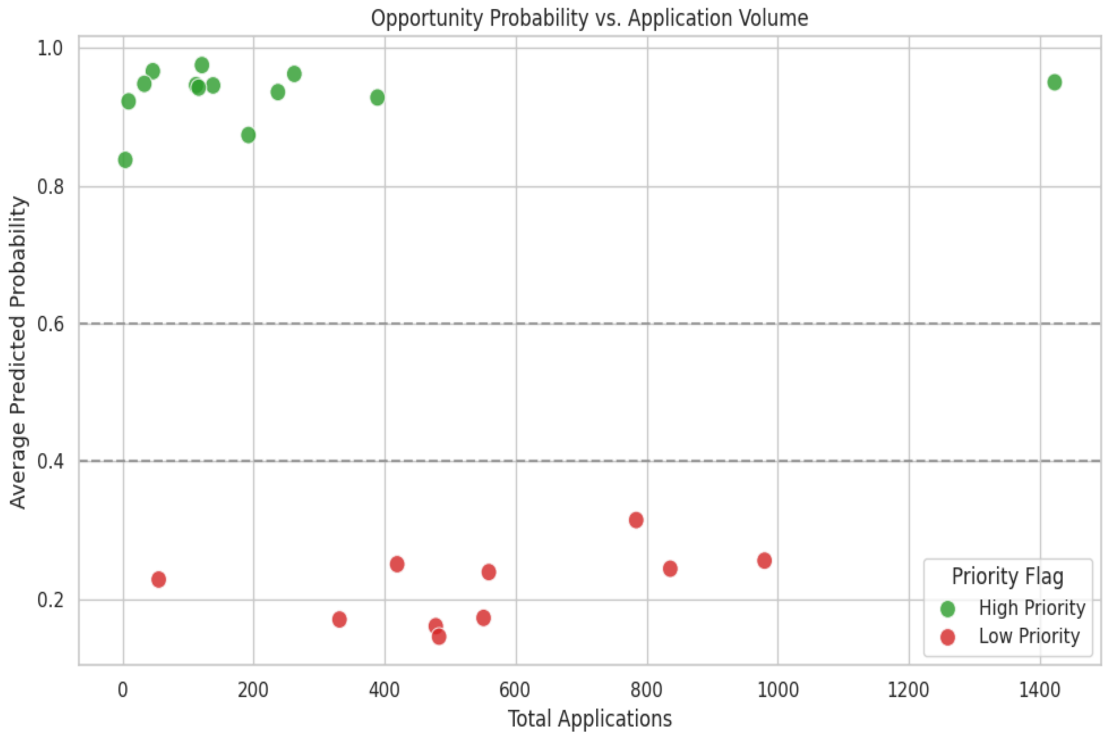

Figure 4: Average predicted probability against total applications, coloured by priority flag.

The chart reveals no simple relationship between raw application volume and predicted participation probability. Several very high-volume opportunities cluster at low predicted probability, while several low-volume opportunities cluster at very high predicted probability. This visual pattern is the first appearance of the category-volume paradox discussed in depth in Section 5.3, and it demonstrates why application count alone is an insufficient signal for prioritisation: a high-volume opportunity is not automatically a high-conversion one.

3.2 Validation and Table Formatting

Before trusting the model's predicted probabilities for planning purposes, they were validated against the simplest possible ground truth: the actual historical active-participation rate observed for each opportunity category in the raw data. Figure 5 places the model's average predicted probability for each category directly alongside the historical actual rate for that same category.

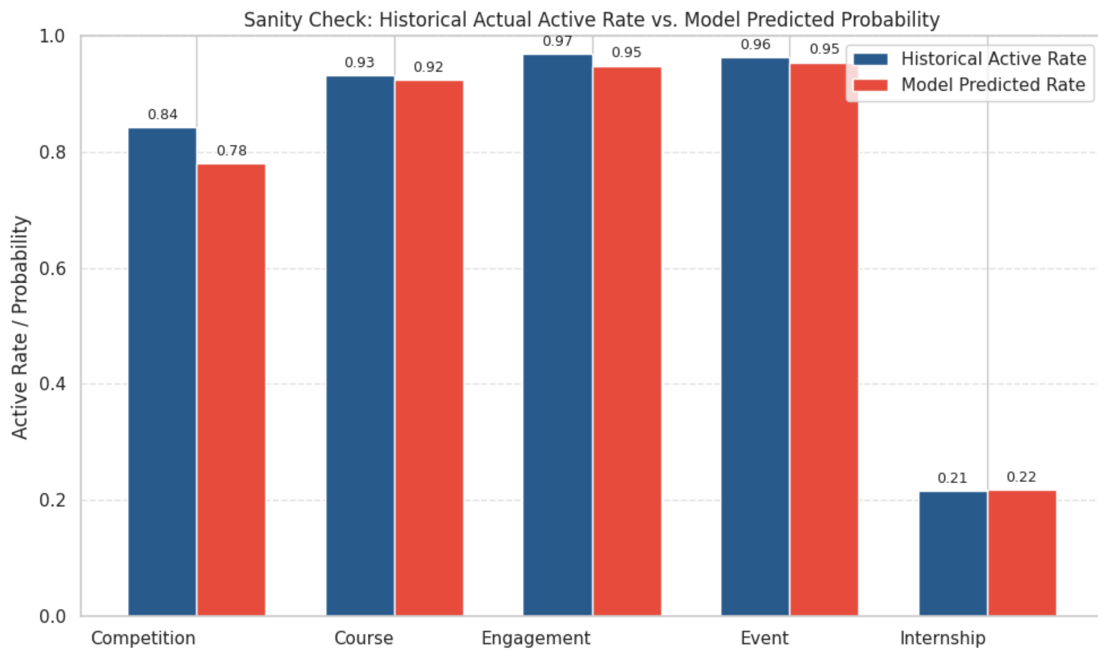

Figure 5: Sanity check comparing historical actual active rate against the model's predicted probability, by category.

The close alignment between the two bars in each category pair confirms that the model's probability outputs are well-calibrated: it is not systematically over- or under-predicting participation for any category relative to what has actually happened historically. Following this validation, each opportunity was assigned a Priority Flag: High Priority for a predicted probability at or above 60%, and Low Priority below that threshold.

3.3 Formatted Probability Table

The validated, flagged probabilities were compiled into a final, ranked table. Table 6 presents the complete ranked list of all 23 opportunities on the platform.

Table 6: Final ranked probability table for all 23 platform opportunities.

#	Opportunity	Category	Predicted Prob.	Priority	Total Applications
1	Join a Student Organisation	Engagement	97.5%	High Priority	121
2	Mental and Physical Health Session	Event	96.6%	High Priority	46
3	Startup Mastery Workshop	Event	96.2%	High Priority	262
4	Career Essentials: Getting Started with Your	Course	95.0%	High Priority	1,423

	Professional Journey				
5	Jump Start: Developing your Emotional Intelligence	Course	94.8%	High Priority	33
6	UX Redesign Challenge	Competition	94.6%	High Priority	112
7	UrbanRenew Challenge	Competition	94.6%	High Priority	138
8	Xperience Design Hackathon	Competition	94.2%	High Priority	116
9	Freelance Mastery Workshop	Event	93.6%	High Priority	237
10	CPR/AED Certification	Course	92.8%	High Priority	389
11	Upload Your First Year Transcript	Engagement	92.2%	High Priority	9
12	Jump Start: Developing your Emotional Intelligence (II)	Course	87.4%	High Priority	192
13	Slide Geeks: A Presentation Design Competition	Competition	83.7%	High Priority	4
14	Health Care Management	Internship	31.5%	Low Priority	784
15	Data Visualization	Internship	25.6%	Low Priority	980
16	Innovation & Entrepreneurship	Internship	25.0%	Low Priority	419
17	Project Management	Internship	24.4%	Low Priority	836
18	Digital Marketing	Internship	23.9%	Low Priority	559
19	AI Ethics Challenge	Competition	22.8%	Low Priority	55
20	Data Visualization Associate	Internship	17.3%	Low Priority	551
21	Digital Strategy Virtual Internship	Internship	17.0%	Low Priority	331
22	Project Management Associate	Internship	16.0%	Low Priority	478

23	Business Consulting	Internship	14.5%	Low Priority	483

Thirteen of the twenty-three opportunities (56.5%) were classified as High Priority, with predicted probabilities ranging from 83.7% to 97.5%. The remaining ten opportunities, all but one of which belong to the Internship category, were classified as Low Priority, with predicted probabilities ranging from 14.5% to 31.5%. This stark, near-total separation between the two priority tiers is explored in full in Section 5.3.

4. Visualizations of Model Outputs

This section presents the remaining model-diagnostic visualisations referenced throughout this report, gathered together for ease of reference: the feature importance ranking, the confusion matrix, the ROC curve, and the two participation-probability heatmaps.

4.1 Feature Importance Chart

Random Forest produces a feature importance score for every input variable, calculated from how much each feature reduces prediction error, averaged across all 300 trees in the ensemble. Figure 6 and Table 7 present this ranking.

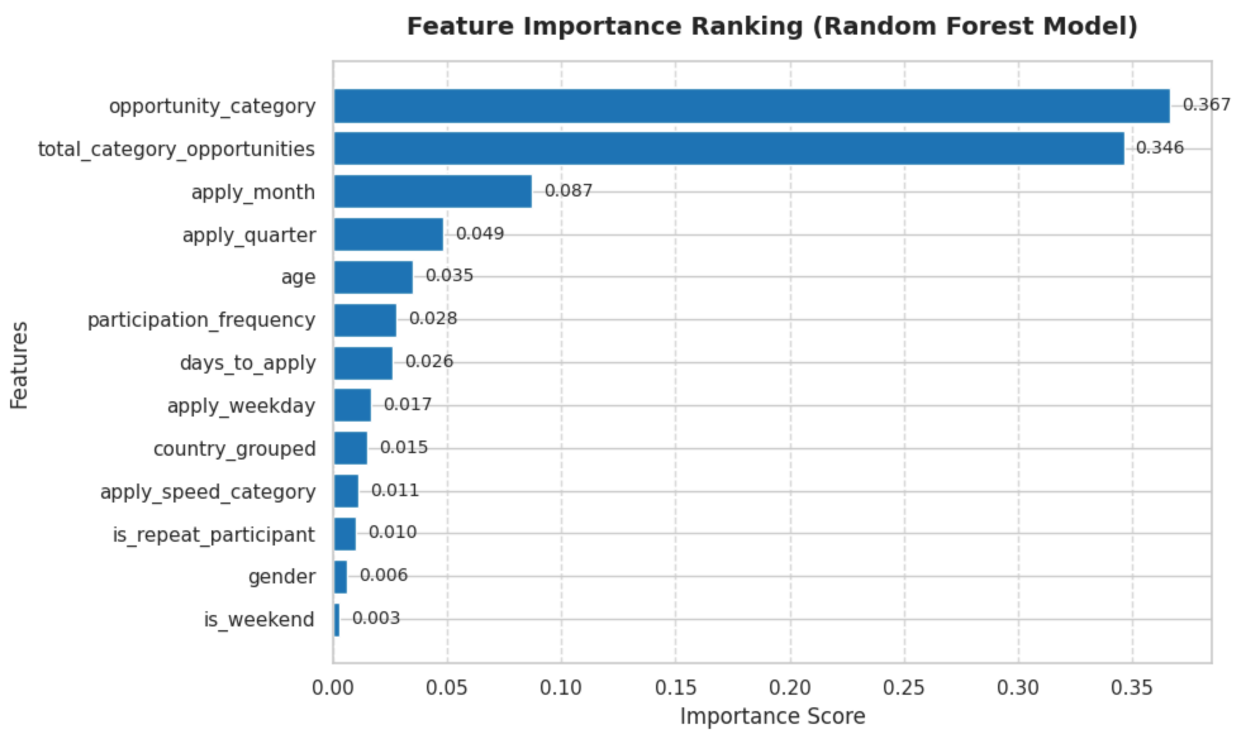

Figure 6: Feature importance ranking from the Random Forest model.

Table 7: Feature importance values, Random Forest model.

Rank	Feature	Importance Score	Share of Total
1	opportunity_category	0.367	36.66%
2	total_category_opportunities	0.346	34.63%
3	apply_month	0.087	8.72%
4	apply_quarter	0.049	4.86%
5	age	0.035	3.50%
6	participation_frequency	0.028	2.77%
7	days_to_apply	0.026	2.60%
8	apply_weekday	0.017	1.68%
9	country_grouped	0.015	1.52%
10	apply_speed_category	0.011	1.13%
11	is_repeat_participant	0.010	1.00%
12	gender	0.006	0.64%
13	is_weekend	0.003	0.29%

The two most important features, opportunity_category and total_category_opportunities, together account for 71.29% of the model's total predictive weight. This is the most consequential finding in the entire analysis: participation is driven overwhelmingly by which category of opportunity a learner applies to, not by who that learner is. Demographic variables are almost irrelevant to the model's decisions by comparison; gender contributes only 0.64% and is_weekend only 0.29%. Application timing (apply_month and apply_quarter) forms a smaller but meaningful secondary cluster at a combined 13.58%.

4.2 Confusion Matrix

While Table 4 describes overall performance, the confusion matrix in Figure 8 breaks the 1,712 test predictions down into the four possible outcomes, which is essential for understanding exactly where and how the model makes errors.

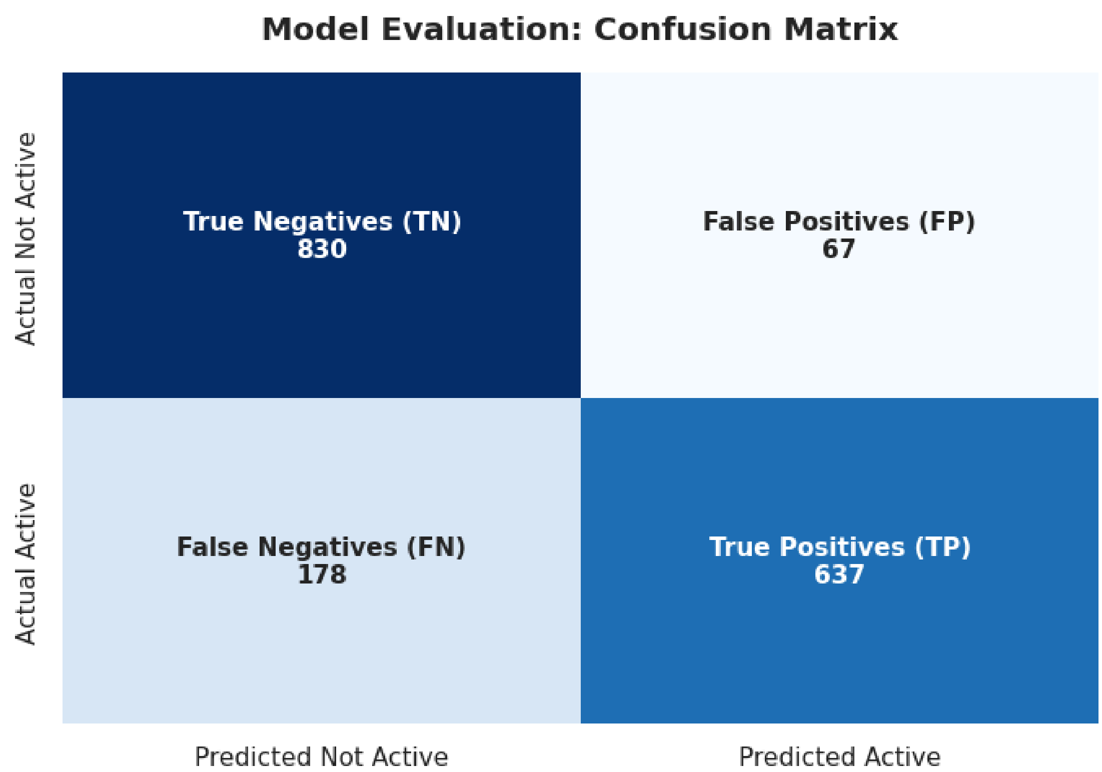

Figure 7: Confusion matrix comparing predicted versus actual participation on the test set.

Of the 1,712 test-set learners, 830 non-active learners were correctly identified as non-active (true negatives) and 637 active learners were correctly identified as active (true positives). The model incorrectly labelled 67 non-active learners as active (false positives) and missed 178 learners who did go on to participate (false negatives). The low false-positive count is what drives the model's strong 90.48% precision score, while the larger false-negative count explains the comparatively lower 78.16% recall.

4.3 ROC Curve

The Receiver Operating Characteristic (ROC) curve in Figure 8 evaluates the model's ability to distinguish between active and non-active learners across every possible probability threshold, not just the default 0.5 cutoff used to produce the confusion matrix above.

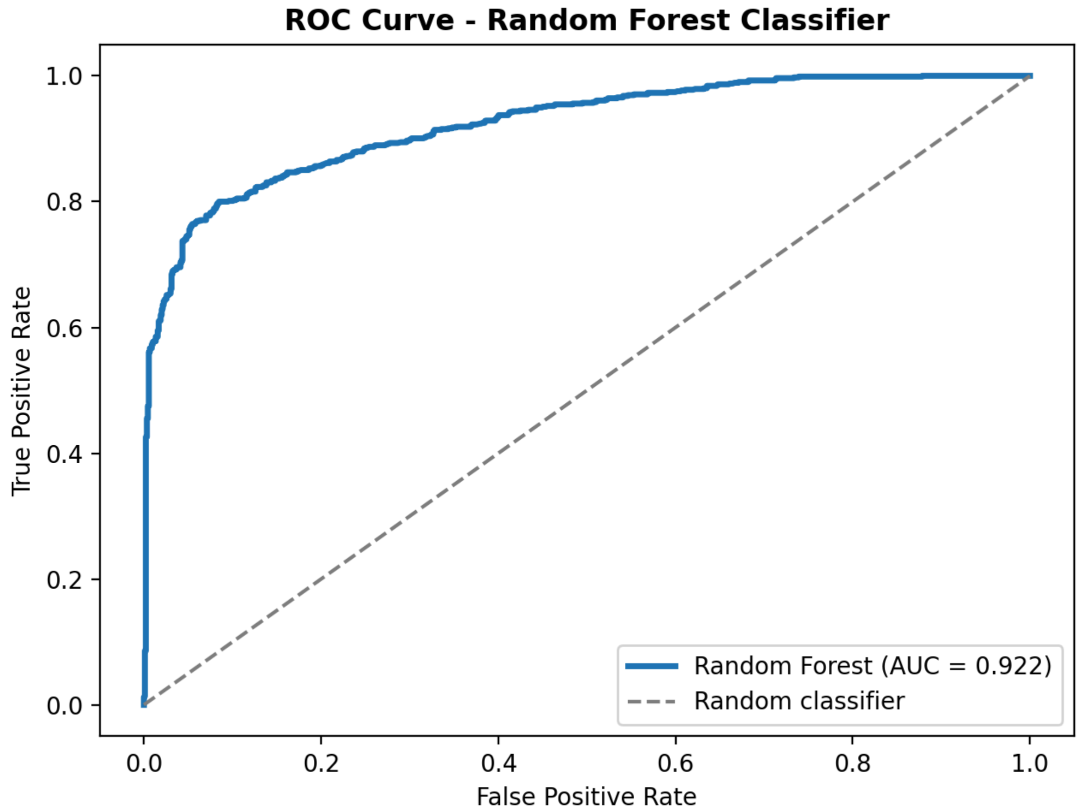

Figure 8: ROC curve for the Random Forest classifier, showing a ROC-AUC of 0.9225.

The curve bends strongly toward the top-left corner and remains well above the random classifier's diagonal line, confirming an AUC of 0.9225. This means that when comparing one active and one non-active learner, the model correctly assigns a higher probability to the active learner 92.25% of the time.

4.4 Heatmaps

To make the probability table in Section 3.3 easier to interpret, two complementary heatmaps were created. Figure 9 summarizes the full probability table into a category-level view.

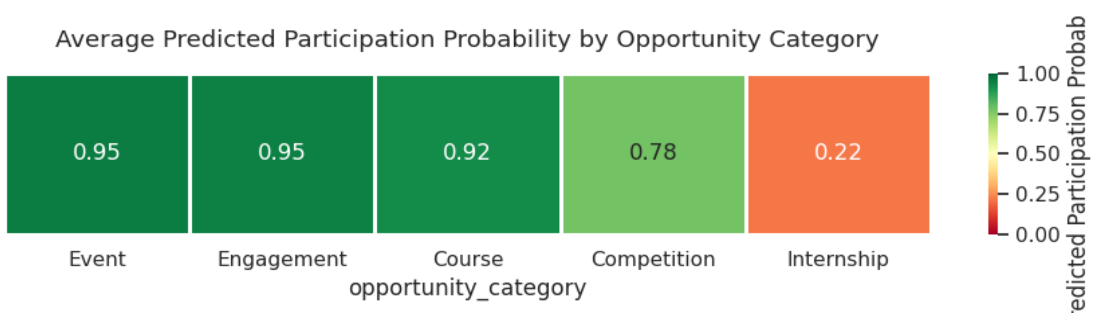

Figure 9: Average predicted participation probability by opportunity category.

This heatmap uses a red-to-green colour scale to distinguish low- and high-probability categories. The Internship category stands out clearly in red compared to the green-dominated categories. Figure 11 provides a more detailed opportunity-level view.

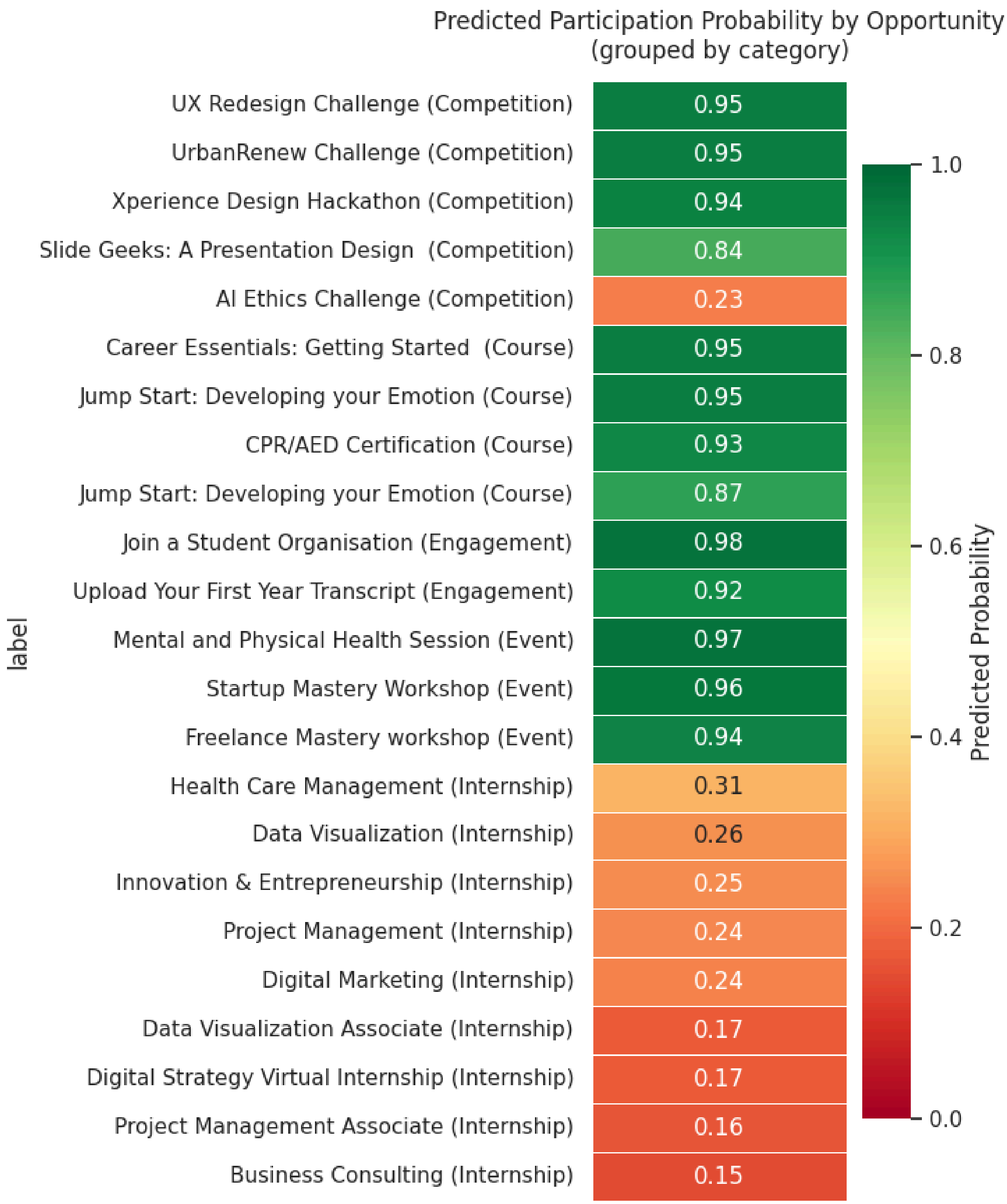

Figure 11: Predicted participation probability for every opportunity, grouped by category.

This second heatmap lists every individual opportunity, grouped by category and sorted by descending probability within each group, allowing a programme manager to identify not just which category needs attention, but exactly which named opportunities within that category are furthest from their historical peers.

5. Interpretation of Findings

5.1 Overall Model Performance

The Random Forest model reaches 85.69 percent test accuracy, an F1-score of 0.8387, and a 0.9225 ROC-AUC score, which together confirm it is highly reliable at telling which applicants are likely to actively participate and which are more likely to drop out or stay inactive. The narrow 1.43 percentage point gap between training and testing accuracy shows the model is well regularised rather than overfitted. Just as importantly, the pattern of feature importance shows that participation is driven far more by the structure of a programme, the type of opportunity and how many similar options are on offer, than by any individual demographic trait such as gender or country.

5.2 Top Factors Driving Participation

The table below ranks the input features by how much weight the model places on each one when making its predictions.

Feature	Importance Score	Driver Type
Opportunity Category	36.66%	Dominant Structural Driver
Total Category Opportunities	34.63%	Dominant Structural Driver
Apply Month	8.72%	Temporal Driver
Apply Quarter	4.86%	Temporal Driver
Age	3.50%	Demographic / Behavioural Driver
Participation Frequency	2.77%	Behavioural Driver
Days to Apply	2.60%	Temporal Driver
Gender	0.64%	Demographic Driver

Opportunity category alone accounts for 36.66 percent of the model's predictive weight, meaning the type of programme a learner applies to is the single biggest factor in whether they go on to actively participate. Total category opportunities, essentially how many similar options exist within that category, adds a further 34.63 percent, so together these two structural factors explain over 71 percent of what drives the model's predictions. When learners have plenty of appealing options within a category they respond to, participation momentum builds; when a category is thin on choice or poorly structured, engagement suffers regardless of who the learner is.

Application timing adds a meaningful secondary layer: apply month and apply quarter together account for 13.58 percent of the model's weight, showing that academic cycles and seasonal windows genuinely shape whether an applicant follows through. Learner age and participation frequency contribute a smaller but still useful 6.27 percent combined, while gender contributes almost nothing (0.64 percent), confirming that participation is overwhelmingly a story about programme design and timing rather than who the learner is.

5.3 Behavioural Patterns and Strategic Recommendations for Week 4

Reading the feature importance results alongside the probability heatmaps and confusion matrix surfaces a few clear behavioural patterns, together with practical actions the team can carry into Week 4.

The Category Volume Paradox

Event (0.95), Engagement (0.95) and Course (0.92) opportunities achieve near-perfect predicted participation. Internship, despite being the platform's highest-volume category by far, averages a predicted probability of only 0.22, consistently low across all nine individual Internship offerings (each between 0.15 and 0.31). This points to a genuine behavioural pattern rather than a data quirk: high availability without a structured, low-friction commitment leads to steep learner drop-off. Learners consistently prefer short, high-value, interactive formats over long, higher-friction commitments.

Programme Structure Outweighs Demographics

Structural factors, opportunity category and total category opportunities, account for 71.29 percent of the model's total predictive power, while demographic variables such as gender contribute a negligible 0.64 percent. In other words, active participation is driven by how opportunities are designed and how many good options a learner is offered, not by who that learner is. Giving learners a diverse set of choices within already high-performing categories is likely to build further participation momentum.

Seasonality and Application Timing

Apply month, apply quarter and days to apply together form a meaningful secondary cluster of predictive weight (around 16 percent combined). Learner intent clearly shifts across academic quarters and seasonal cycles, and applications submitted during peak alignment windows are noticeably more likely to convert into active participation.

Classification Trade-offs and Silent Engagers

The confusion matrix shows the model is precise (90.5 percent, with only 67 false positives) but conservative: it misses 178 genuinely active learners, misclassifying them as inactive. This conservatism appears linked to low historical activity, learners with a thin participation history can be undercounted even when they go on to actively participate, which is worth factoring into how the model's output is used operationally.

Recommendations for Week 4

- Modularise Low-Performing Categories: restructure the Internship pipeline into smaller, milestone-driven modules or micro-internships to mirror the low-friction, high-engagement structure that works so well for Courses and Events.
- Time-Phased Campaign Scheduling: launch major opportunities in the peak academic and seasonal windows identified by the apply month and apply quarter signals, to avoid off-peak launch decay.
- Target Interventions by Probability Threshold: use the probability rankings in Section 3 to flag opportunities scoring below 0.50, particularly variable categories like Competition (which spans 0.23 to 0.95), and trigger promotional nudges or onboarding support ahead of launch.
- Re-engage Likely False Negatives: build lightweight onboarding flows for first-time or low-frequency applicants, to capture the roughly 178 genuinely active learners the model currently under-predicts.

Dimension	Key Metric	Behavioural Finding	Strategic Action
Internships	Avg. probability: 0.22	High friction / volume paradox	Break into micro-internships
Events & Courses	Avg. probability: 0.92–0.95	Preferred low-friction formats	Scale and replicate structural model
Timing Drivers	13.58% feature importance	Strong seasonal dependency	Align launches with peak academic months
Error Matrix	178 false negatives	Conservative prediction baseline	Target low-frequency active applicants

6. AI-Assisted Analysis

In line with the project's Action Points, an AI assistant was used throughout Week 3 as a collaborative tool for code review, debugging, and narrative structuring, under continuous human direction and validation from the team.

AI Tools Used

- Large language model assistants (used for prompt-assisted exploratory analysis, code structuring and drafting support).
- Python data science stack: pandas, scikit-learn, matplotlib and seaborn.
- Jupyter Notebook, used to run, check and debug every generated script before it was accepted.

Prompts Used

The following prompt structures guided the AI-assisted parts of the work, to keep the process transparent and repeatable:

- Strategic validation and model feature hierarchy: requesting help turning the team's own Random Forest results and feature importances into a clean, human-readable priority matrix and narrative breakdown that non-technical stakeholders could follow without reading raw code output.
- Calibration review and executive chart styling: requesting Matplotlib and Seaborn code to visualise the calibration comparison between predicted probabilities and actual historical participation rates, with a cohesive colour palette and clear data labels suitable for a presentation slide.
- Diagnostic visualization and dashboard layout: requesting a polished, presentation-ready layout for the evaluation metrics and confusion matrix, including a readable colour scheme and an automated warning tag if the train-test accuracy gap exceeded 0.07.
- Executive synthesis and strategic recommendations: requesting help synthesising the feature importance results and probability heatmaps into an executive-level summary, explaining why programme structure so heavily outweighs demographic factors, and refining three to four actionable recommendations for Week 4.

Insights Generated

- Feature importance analysis pointed to opportunity category and category-level opportunity density as the dominant predictors of active participation, together explaining over 71 percent of the model's predictive weight.
- Opportunities were grouped into High, Medium and Low priority bands based on predicted probability, giving a ready-made ranking for outreach planning.
- Category-level comparisons showed the model's predicted probabilities closely track real historical participation baselines, supporting confidence in the estimates.

Validation Process

Every AI-assisted output was checked by the team before being accepted into the final analysis:

- Logic and calculation checks: all model outputs and data transformations were independently reproduced using standard Python workflows.

- Sanity checks against historical baselines: predicted probabilities were compared category by category against real historical participation rates.
- Overfitting and generalisation testing: a stratified train-test split was used, and the accuracy gap was monitored to stay below a 7 percent threshold.
- Code and column verification: generated scripts were reviewed line by line to confirm variable and column names matched the project's actual schema.

7. Conclusion

This report has documented the complete Week 3 pipeline, from re-verifying the Week 2 dataset through to a fully evaluated, benchmarked, and interpreted Random Forest classifier. The model achieves strong, well-validated performance (85.69% accuracy, 0.9225 ROC-AUC), was selected over three credible alternative algorithms with quantitative justification, and its outputs were cross-checked against historical ground truth before being trusted for planning purposes.

Most importantly, the analysis surfaced a specific, actionable, and genuinely non-obvious finding: application volume and participation probability are not the same thing, and the platform's highest-volume category (Internship) is simultaneously its lowest-converting one. This single insight, together with the ranked probability table in Section 8.4 and the recommendations in Section 10.5, gives the team a concrete, evidence-based foundation on which to build the Week 4 final report and executive presentation.

8. Additional Resources

The table below lists the supporting files that accompany this report.

Table 8: Supporting notebooks and data files.

Resource Type	Description	Link
Model Training Notebook	Complete notebook containing preprocessing, feature engineering, model development, and evaluation.	https://colab.research.google.com/drive/1DSagBc2Do5V8GI6k1Iv5xGfNLiR1fd09?usp=sharing
Model Comparison Notebook	Notebook containing the comparative evaluation of all four implemented machine learning models.	https://drive.google.com/file/d/1Yu-eutZ8boN2Nux2apRv68nq1YV1QsGg/view?usp=sharing
Processed Dataset	Week 2 preprocessed dataset used to train and test the model (8,558 records).	https://drive.google.com/file/d/1z_NW93p3Hr-9AQtafDN4nnPXbdh1iq-J/view?usp=sharing

Resource Type	Description	Link
Trained Model	Saved Random Forest model, ready to be reloaded for future predictions.	https://drive.google.com/file/d/1LY9MkhujuDFP5vFgSnymYqY1_QQ82BL7/view?usp=sharing
Encoders	Saved label encoders used to convert categorical fields into the format the model expects.	https://drive.google.com/file/d/1Zx2fT6z_gOZ4FGqRwy6sDvHSOLkf_I3p/view?usp=sharing
Metrics	Saved evaluation metrics: accuracy, precision, recall, F1-score, ROC-AUC and confusion matrix.	https://drive.google.com/file/d/1kcoU2SpTnx2QsmR2odPtT5UCf7TOgoYY/view?usp=sharing
Opportunities Probabilities	Contains the predicted participation probabilities for each opportunity category generated by the Random Forest model.	https://drive.google.com/file/d/1ozV1Sv3nOW4XSQvLSx11KEHg2zic6L-Y/view?usp=sharing
Formatted Probability Table	Final ranked probability table with priority flags for every opportunity.	https://drive.google.com/file/d/1hGfFHwjin2pN94pjMYQA8m69d2dfb5mQ/view?usp=sharing